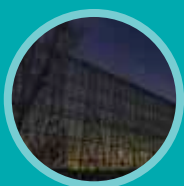


SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Earth, we learn all about wildfires, in Space Age, we marvel at the first pictures taken from the James Webb Space Telescope and in Health, we learn about an odd phenomenon known as Phantom Ringing.



Alien messages coming soon

A series of new calculations done as a part of a study by researchers Arjun Berera and Jaime Calderón-Figueroa suggest that aliens could send quantum messages to Earth. During their study they examined "possibility of achieving quantum communication using photons across interstellar distances". <https://dgit.in/alienmessages>

WHAT'S NEW

First electron whirlpools

Researchers at the Massachusetts Institute of Technology and Weizmann have finally been successful in discovering electron whirlpools. The presence of electron whirlpools had previously been predicted; however, it had never been confirmed, until now. Commenting on this finding, Leonid Levitov, a professor of physics at MIT, said, "Electron vortices are expected in theory, but there's been no direct proof, and seeing is believing." He added, "Now we've seen it, and it's a clear signature of being in this new regime, where electrons behave as a fluid, not as individual particles." Previously, the closest scientists had gotten to discovering such behaviour was in the year 2017. A team of researchers from the University of Manchester



reported that they were successful in observing fluid-like electron behaviour in graphene. To get a visual representa-

tion of the electron vortices, the researchers observed tungsten ditelluride (WTe₂). They synthesised pure single crystals of the same. After that, structured sampling was done, and current flow was observed in the material using special equipment. The findings brought out experimental evidence that could now be "leveraged for low-power next-generation electronics".

<https://dgit.in/electronwhirlpools>

The reason behind tiny arms of dinosaurs found

A newly discovered meat-eating dinosaur named *Meraxes gigas* has been able to provide clues helping answer the age-old question – why did dinosaurs have tiny arms? This dinosaur could provide further clues about the evolution and biological structure of *Carcharodontosaurus* and *Tyrannosaurus rex*. This is because it bears a resemblance to the famous carnivorous dinosaur. Commenting on this, a member of



the research team said that this dinosaur's body plan was "surprisingly similar" to that of *T. rex*. He added, "But, they're not particularly closely related to *T. rex*. So, having this new discovery

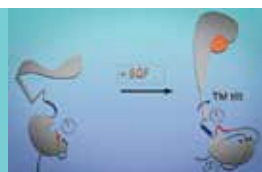
allowed us to probe the question of, 'Why do these meat-eating dinosaurs get so big and have dinky little arms?'" Statistical data found from the discovered remains showed that the growth patterns in all the three families of theropods were almost the same. The study found that *Meraxes* and the related dinosaurs underwent rapid evolution. However, it did not take long for them to go extinct too.

<https://dgit.in/dinoarms>



First fires

Researchers, using advanced AI techniques, have finally been able to uncover the first evidence of the use of fire. These date back to at least 80,000 years and stand as one of the oldest known pieces of evidence of the use of fire. <https://dgit.in/ancientfire>



Shifting shapes, aiding growth

MIT chemists have uncovered new insights into the functioning and influence of the EGF receptor in the human body. These discoveries about the EGFR could potentially help in the development of more effective cancer medication. <https://dgit.in/shiftingshapes>



Making artificial hearts

A team of bioengineers from the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS) have made great leaps towards developing effective bio-fabrication techniques, solving a 300-year-old mystery in the process. <https://dgit.in/artificialhearts>